ISLR Exercises: Homework 6

Haresh Sambandamoorthy | DSCI 552 | Homework 6

2. ISLR 6.6.3 - Lasso constraint

The allowed total absolute size of the coefficients is increased (s gets larger), giving the model more freedom.

Quantity	Choice	Why
(a) Training RSS	iv - Decreases	A model with more freedom can fit training data at least as well, or equally well.
(b) Test RSS	ii - U-shaped	Some flexibility helps at first; too much may fit noise.
(c) Variance	iii - Increases	A flexible fit can change more when the training data changes.
(d) Squared bias	iv - Decreases	The fit can capture more of the true pattern.
(e) Irreducible error	v - Constant	Unpredictable noise is not changed by s.

These are the usual broad patterns intended by the exercise; a single real dataset may behave differently.

3. ISLR 6.6.5 - Identical predictors

Let the two predictors be identical, taking the values a and -a across the two observations, with responses b and -b. Define t = beta1 + beta2. The fitted value depends only on t, not on the separate coefficients.

(a) Ridge objective

Minimize $2(b - a\,t)^2 + \text{lambda}(\text{beta1}^2 + \text{beta2}^2)$.

(b) Ridge coefficients

For any fixed t, the squared penalty is smallest when beta1 = beta2 = t/2. The resulting ridge estimates are $\text{beta1} = \text{beta2} = 2ab/(4a^2 + \text{lambda})$, for positive lambda.

(c) Lasso objective

Minimize $2(b - a\,t)^2 + \text{lambda}(|\text{beta1}| + |\text{beta2}|)$.

(d) Why the lasso answers need not be unique

When the two coefficients have the same sign, their absolute values add to |t|. Thus (0,t), (t/2,t/2), and (t,0) all give the same predictions and penalty. The best total is sign(b/a) times $\max(|b/a| - \text{lambda}/(4a^2), 0)$, when a is nonzero. If this total is nonzero, there are infinitely many equally good coefficient pairs. If it is zero with positive lambda, (0,0) is the unique pair.

4. ISLR 8.4.5 - Bagging: votes versus probabilities

Six of the ten trees predict the red class with probability above 0.50, while four predict below 0.50. **Majority voting predicts Red.** The ten probabilities add to 4.50, giving an average of 0.45. **Averaging probabilities predicts Green.** Voting counts the winners only, while averaging uses the confidence of every tree.

5. ISLR 9.7.3 - Maximum-margin separator

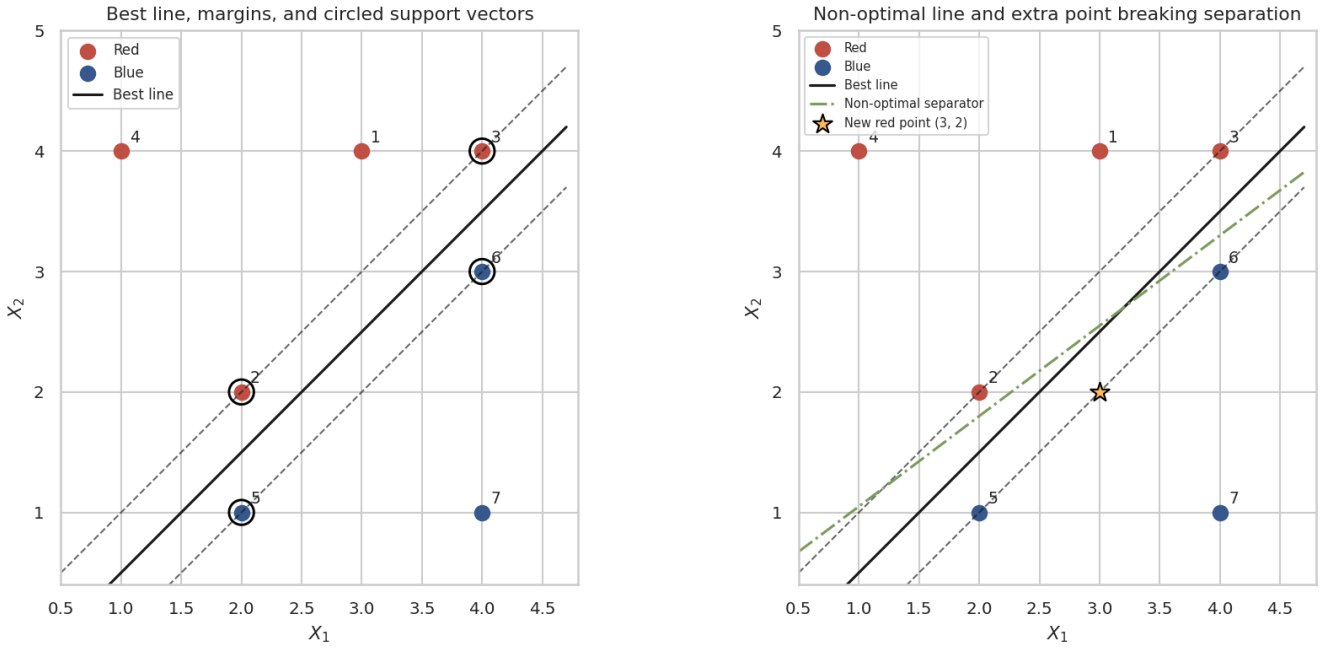


Figure 11. Maximum-margin line (solid), two margins (dashed), alternative line and a new point.

(a) Red points: (3,4), (2,2), (4,4), (1,4). Blue points: (2,1), (4,3), (4,1).

(b) The maximum-margin separating line is X2 = X1 - 0.5, or 0.5 - X1 + X2 = 0.

(c) Predict Red when 0.5 - X1 + X2 is positive, and Blue otherwise. One valid coefficient set is beta0 = 0.5, beta1 = -1, beta2 = 1.

(d) The two margin boundaries are X2 = X1 and X2 = X1 - 1. Each is 0.5/sqrt(2) (about 0.354) from the separating line.

(e) Support vectors: Red points 2 and 3; Blue points 5 and 6. They lie on the two margin boundaries.

(f) Moving point 7 slightly will not change the maximum-margin line, because it does not touch either margin.

(g) The line X2 = 0.75 X1 + 0.30 also separates the original groups but has a smaller margin, so it is not optimal.

(h) Add a Red point at (3,2). This falls on the segment between two Blue points, so no straight line can then separate all Red points from all Blue points.

6. Extra practice (optional)

ISLR 5.4.2 - Bootstrap inclusion chance

(a)-(c) Any one draw misses observation j with probability (n-1)/n. All n independent draws miss it with probability $(1-1/n)^n$; therefore its probability of appearing at least once is $1 - (1-1/n)^n$.

(d) n = 5: 0.67232. (e) n = 100: 0.63397. (f) n = 10,000: 0.63214. (g) The value approaches 1-1/e = 0.63212 as n grows. (h) A simulation of 10,000 bootstrap samples of size 100 produced a proportion of 0.6304, close to the theoretical 0.6340.

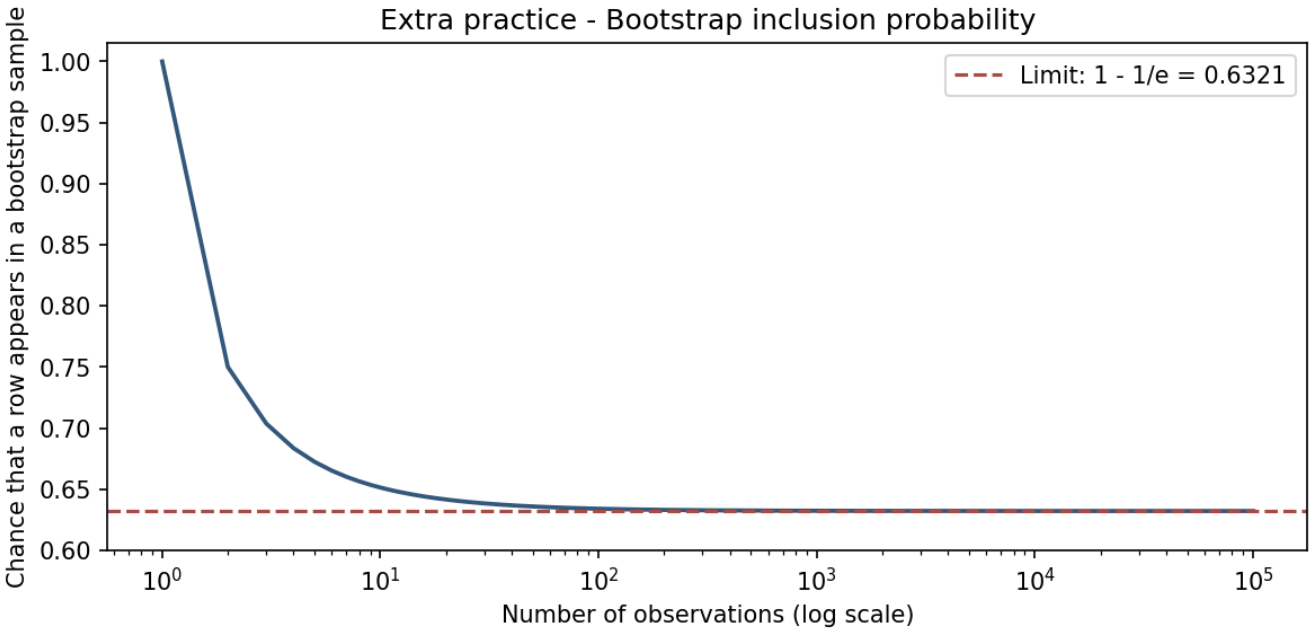


Figure 12. Probability that an observation appears at least once in a bootstrap sample.

ISLR 6.8.4 (first-edition numbering) - Ridge as lambda increases

(a) Training RSS: iii, increases. (b) Test RSS: ii, usually U-shaped. (c) Variance: iv, decreases. (d) Squared bias: iii, increases. (e) Irreducible error: v, remains constant.

ISLR 8.4.4 - Tree partitions

(a) The pictured split puts X1 at least 1 in a region with prediction 5. On the left, a cut at X1 = 0 and then X2 = 1 gives the four remaining terminal values 15, 3, 0, and 10. (b) For the provided tree, first split at X2 = 1. For X2 below 1, split on X1 = 1 into values -1.80 and 0.63. For X2 between 1 and 2, split on X1 = 0 into -1.06 and 0.21. At X2 at least 2, predict 2.49.

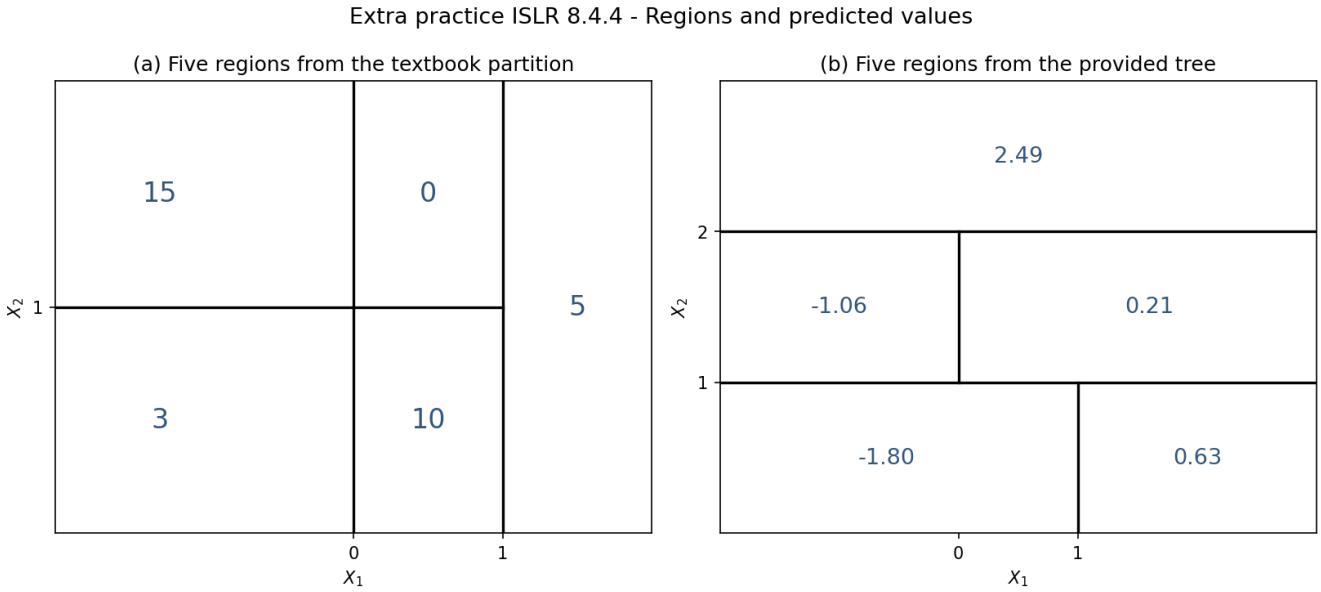


Figure 13. Regions corresponding to the two decision-tree questions.

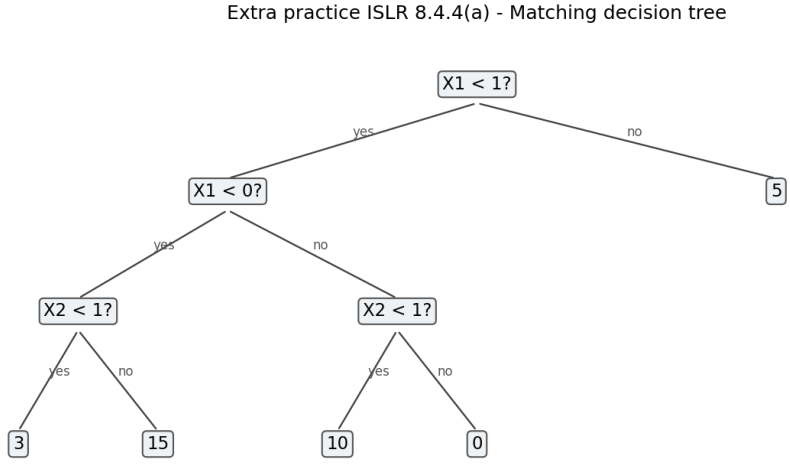


Figure 14. Tree matching the first partition diagram.

ISLR 9.7.2 - Nonlinear decision boundary

(a) $(1 + X1)^2 + (2 - X2)^2 = 4$ is a circle with center (-1,2) and radius 2. (b) Values above 4 are outside; values at most 4 are inside or on the circle. (c) (0,0) Blue; (-1,1) Red; (2,2) Blue; (3,8) Blue. (d) Expanding gives $1 + 2\ X1 + X1^2 - 4\ X2 + X2^2 = 0$, a linear rule in the expanded inputs X1, $X1^2$, X2 and $X2^2$.

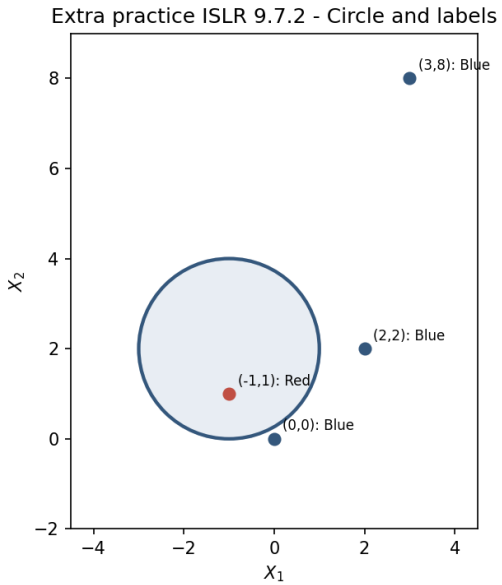


Figure 15. The circular boundary and four example observations.

References

[1] DSCI 552 Homework 6, user-supplied question sheet, and the accompanying APS training and test files.

[2] UCI Machine Learning Repository: APS Failure at Scania Trucks.
https://archive.ics.uci.edu/dataset/421/aps+failure+at+scania+trucks

[3] James, Witten, Hastie and Tibshirani. An Introduction to Statistical Learning, first and second editions.
https://www.statlearning.com/

[4] XGBoost documentation, Parameters and Introduction to Boosted Trees:
https://xgboost.readthedocs.io/en/stable/parameter.html and
https://xgboost.readthedocs.io/en/stable/tutorials/model.html

[5] scikit-learn documentation, SGDClassifier:
https://scikit-learn.org/stable/modules/generated/sklearn.linear_model.SGDClassifier.html

AI use acknowledgment

AI tools were used to assist with organizing the code, compiling the report, and checking formatting. All reported numerical results were generated from the included APS datasets using the accompanying reproducible code.